

Multimodal Depression Detection from Speech and Linguistic Analysis Across Conversational Phases

A Project Report

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**Faculty of Engineering and Technology, Institute of Technical Education and
Research**

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE) UNIVERSITY

Bhubaneswar, Odisha, India (June 2026)



CERTIFICATE

This is to certify that the project report titled “**Multimodal Depression Detection from Speech and Linguistic Analysis Across Conversational Phases**” being submitted by Debi Prasad Nanda, Chandna Nayak, Prabyasachi Mohanty, Debasish Nanda, Pushpaja Priyadarshini of Section 2241033 to the Institute of Technical Education and Research, Siksha ‘O’ Anusandhan (Deemed to be) University, Bhubaneswar for the partial fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering is a record of original confide work carried out by them under my/our supervision and guidance. The project work, in my/our opinion, has reached the requisite standard fulfilling the requirements for the degree of Bachelor of Technology. The results contained in this project work have not been submitted in part or full to any other University or Institute for the award of any degree or diploma.

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Date:

Signature of Students

i.

ii.

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DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/fact/source in our submission. We understand that any violation of the above will cause for disciplinary action by the University and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

Signature of Students with Registration Numbers

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iii.

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Date:

REPORT APPROVAL

This project report titled “**Multimodal Depression Detection from Speech and Linguistic Analysis Across Conversational Phases**” being submitted by Debi Prasad Nanda, Chandna Nayak, Prabyasachi Mohanty, Debasish Nanda, Pushpaja Priyadarshini approved for the degree of Bachelor of Technology in Computer Science and Engineering.

Examiner(s)

Supervisor

Project Coordinator

PREFACE

Depression is a prevalent mental health condition whose early identification remains challenging in resource-limited settings. Clinical interviews capture both how a person speaks and what they say; together these modalities encode behavioural cues relevant to affective state. This project develops an automated screening pipeline that processes interview audio and transcripts from a DAIC-WOZ style corpus, removes interviewer speech, extracts acoustic and linguistic features, and evaluates machine learning classifiers for binary depression detection (PHQ-8 binary label). Conversations are further divided into three temporal phases that are early, middle, and late; to study how discriminative patterns evolve over time. Experiments compare audio-only, text-only, and fused multimodal representations using Random Forest, XGBoost, SVM, Logistic Regression, CatBoost, and a feed-forward neural network, with stratified hold-out and k-fold validation. Text-centric models (CatBoost, Phase 1 accuracy 85.56%) outperform audio-only baselines (best 67.62%), while multimodal Random Forest achieves strong early-phase performance (83.89% accuracy, F1 0.70) with complementary acoustic cues. Results indicate that preprocessing quality, class-imbalance handling, and conversational phase strongly influence detection performance.

INDIVIDUAL CONTRIBUTIONS

Team member	Contributions
Debi Prasad Nanda	project workflow design; phase-wise conversational segmentation design; audio and multimodal feature extraction; methodology planning; experimental analysis and interpretation.
Chandna Nayak	Literature survey; text feature extraction; transcript-related analysis; documentation and report preparation.
Prabyasachi Mohanty	Machine learning model implementation; coding and experimentation; model training and evaluation; result generation and validation.
Debasish Nanda	Audio preprocessing support; preprocessing pipeline implementation; dataset preparation; experimentation support; documentation.
Pushpaja Priyadarsini	Literature survey; preprocessing support; dataset organization; documentation and report formatting

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1. INTRODUCTION

1.1 Project Overview

Mental health disorders, particularly depression, affect millions worldwide and are frequently under-diagnosed. Behavioural analysis of spontaneous speech during structured clinical interviews offers a non-invasive signal for screening. This project implements a complete experimental pipeline: acquisition of paired audio–transcript sessions, rigorous preprocessing, phase-wise segmentation of participant responses, extraction of acoustic and linguistic descriptors, multimodal fusion, and comparative evaluation of classical machine learning models. The objective is not clinical diagnosis but reproducible research into automated cues that correlate with PHQ-8 binary depression labels on a public-style interview corpus.

1.2 Motivation

Traditional screening relies on self-report questionnaires and clinician time, both of which scale poorly. Speech carries prosodic and temporal markers (pause patterns, energy, pitch variability) while language carries semantic and lexical markers (negation, uncertainty, first-person focus). Multimodal systems can exploit complementary information. Furthermore, affective expression may change as an interview progresses; phase-wise analysis tests whether early conversational behaviour is more or less informative than later behaviour.

1.3 Objectives

To design audio and transcript preprocessing suitable for participant-only behavioural modelling; to extract reproducible acoustic (MFCC, spectral, prosodic) and text (lexical, pause, SBERT) features; to segment each interview into three conversational phases; to train and compare multiple classifiers under stratified validation with imbalance-aware decision thresholds; and to analyse modality-wise and phase-wise performance with standard classification metrics.

1.4 Uniqueness of Work

The work integrates DAIC-WOZ-style multimodal processing with explicit phase-wise temporal analysis and a broad classifier comparison (including CatBoost and a shallow

neural network) on a consistently preprocessed subset of 141 participants. A fixed decision threshold (0.40) tuned for depressed-class recall is applied uniformly across models, enabling fair comparison under class imbalance.

1.5 Scope of Work

Scope is limited to offline batch processing of English clinical interviews, binary PHQ-8 labels, and classical ML models. Real-time deployment, multilingual speech, and fine-grained PHQ-8 severity regression are out of scope.

1.6 Applications

Potential applications include decision-support for tele-health screening, longitudinal monitoring research, and educational tools for computational psychiatry—always subject to ethical review and never as standalone diagnosis.

1.7 Report Layout

Chapter 2 reviews related work. Chapter 3 details methodology. Chapter 4 presents results. Chapter 5 concludes. References, appendices, team reflection, and similarity report follow.

Major depressive disorder affects mood, cognition, and somatic function, with global prevalence exceeding hundreds of millions of cases. Screening instruments such as the Patient Health Questionnaire (PHQ-8) reduce administrative burden but still require participant honesty and clinician availability. Automated analysis of behavioural signals recorded during structured interviews does not replace clinical judgement; rather, it offers a secondary evidence stream that may prioritise follow-up in digital mental health services.

Speech production involves respiration, phonation, and articulation subsystems coordinated by cognitive and affective circuits. Depression-related psychomotor retardation can slow articulation rate and increase pause frequency without necessarily altering vocabulary. Conversely, cognitive distortions may appear primarily in lexical choice while gross motor speech remains intact. A multimodal framework therefore aligns with neuropsychological models in which affective illness has both motor and cognitive signatures.

The virtual interviewer paradigm standardises stimulus delivery, reducing interviewer-variance that complicates multi-site corpora. Participants respond to empathic prompts, free-recall autobiographical segments, and symptom-oriented questions. Because Ellie is implemented as a scripted agent, transcripts identify speaker roles reliably, enabling automated removal of non-participant speech a prerequisite for fair acoustic modelling.

Binary PHQ-8 labels simplify evaluation but collapse severity gradations. The project respects this constraint to remain comparable with published DAIC-WOZ baselines while acknowledging that regression toward continuous scores is a natural extension. Class imbalance (30% positive) mirrors realistic screening populations where most attendees test negative; metrics must be interpreted accordingly.

Ethical deployment of depression detection systems demands informed consent, transparency about model limitations, and human oversight. The present work is positioned explicitly as research software without regulatory clearance for diagnostic use. Privacy protections for audio and transcript data include de-identified participant IDs and local processing without cloud transmission of raw sessions in the final pipeline configuration.

From an engineering perspective, the project exercises the full machine learning lifecycle: data auditing, cleaning, feature design, training, validation, error analysis, and documentation. Such breadth distinguishes a final-year research project from a single-algorithm experiment and prepares graduates for industry ML pipelines subject to audit and reproducibility requirements.

2. LITERATURE SURVEY

2.1 Review of Related Works

Automated depression detection from interviews has been studied extensively on the DAIC-WOZ corpus. Systems range from audio spectral features with SVMs to deep sequential models on raw waveforms and transformer-based text encoders.

2.1.1 Audio-Based Depression Detection

Acoustic studies emphasize low energy, reduced pitch variation, increased pause duration, and jitter/shimmer anomalies. MFCC and prosodic features remain strong baselines when data are limited.

2.1.2 Text-Based Depression Detection

Linguistic markers include increased first-person pronouns, negation, sadness-related vocabulary, and reduced semantic coherence. Pre-trained sentence embeddings (e.g., SBERT) capture contextual semantics beyond bag-of-words models.

2.1.3 Machine Learning Techniques Used in Prior Work

Prior work employs SVMs, random forests, gradient boosting, and increasingly deep networks. For small clinical datasets, regularized linear models and ensembles often generalize better than very deep architectures.

2.2 Research Gaps Identified

Many studies pool entire interviews, ignoring intra-session dynamics. Multimodal fusion strategies are not always comparable across papers. Class imbalance is often handled inconsistently, inflating accuracy while harming recall for the depressed class.

2.3 Problem Identification

This project addresses: (i) reproducible participant-only preprocessing, (ii) explicit early/middle/late phase modelling, and (iii) systematic comparison of six model families under shared validation and thresholding.

Table 2.1. Literature survey comparison.

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
1	Guo et al. (2022)	Feature Extraction: Transformer-based (RoBERTa, wav2vec 2.0) with Topic Attention Classification: Two-branch multimodal transformer with late fusion	DAIC-WOZ dataset	Accuracy: 90%+ (reported improvement) Recall: Improved via attention	Effective use of topic-level attention improves detection; handles data scarcity using pre-trained models but still limited by small dataset size
2	Gan et al. (2025)	Feature Extraction: LLM (LLaMA), Audio processing Classification: Teacher-Student multimodal fusion with attention	DAIC-WOZ dataset	F1 Score: 0.991	Very high accuracy using teacher-student architecture; strong multimodal fusion but computationally complex and data-dependent
3	Gong & Poellabauer (2017)	Feature Extraction: Topic Modeling (context-aware segmentation) Classification: Multimodal feature vector + feature selection	DAIC-WOZ dataset	Improved performance over baseline methods	Captures temporal/contextual details effectively; limited by small sample size and feature dimensionality issues
4	Nykoniuk et al. (2025)	Feature Extraction: CNN + Bi-LSTM + Self-Attention Classification: Early and Late Fusion models	DAIC-WOZ, EDAIC-WOZ datasets	Accuracy: 86% F1 Score: 0.79	Early fusion performs better; multimodal improves accuracy but dataset imbalance affects reliability
5	Yoo & Oh (2023)	Feature Extraction: CNN (audio), Transformer (text) Classification: Tensor Fusion Network + LSTM	DAIC-WOZ dataset	Moderate improvement (no exact metrics provided)	Fusion of text and voice improves detection; lacks detailed quantitative evaluation and scalability analysis
6	Xu et al. (2025)	Feature Extraction: BERT (text), wav2vec 2.0 (audio) Classification: Bi-LSTM + multi-scale convolution + adaptive pooling	CMDC & DAIC-WOZ datasets	F1 Score: Improved significantly RMSE reduced	Strong cross-modal feature integration; handles multiple datasets but requires large training data
7	Park & Moon (2022)	Feature Extraction: BERT-CNN (text), CNN-BiLSTM (audio) Classification: Attention-based multimodal fusion	DAIC-WOZ dataset	Improved accuracy over single-modal models	Attention-based fusion improves accuracy; still dependent on quality of preprocessing and multimodal alignment

Depression manifests in measurable changes to motor-speech programs and language production. Reduced affect often correlates with flatter pitch contours, longer pauses, and lower speech rate. Computational psychiatry has therefore treated the clinical interview as a structured probe that elicits both linguistic content and paralinguistic form. The DAIC-WOZ corpus standardised this setting by providing aligned audio, transcripts, and PHQ-8 scores for hundreds of participants interviewed by a virtual agent [1].

Audio-based systems historically relied on hand-crafted descriptors exported from openSMILE or custom Librosa pipelines. Mel-frequency cepstral coefficients compress spectral envelopes into compact coefficients that correlate with vocal tract configuration. Spectral centroid and bandwidth describe energy distribution across frequency bins, while zero-crossing rate distinguishes voiced and unvoiced regions. Prosodic features such as pitch mean and variance capture monotonicity often reported in depressed speech. Voice-quality measures including jitter and shimmer quantify micro-perturbations in periodicity linked to neuromotor control. Recent multimodal studies on DAIC-WOZ continue to incorporate such handcrafted acoustic features alongside deep representations for depression classification [2], [3].

Text-based approaches moved from bag-of-words and LIWC categories toward distributed representations. Transformer-era sentence encoders allow a single vector to summarise utterance-level semantics while remaining computationally feasible on consumer hardware. In parallel, classical lexical features—negation frequency, first-person pronoun density, and type-token ratio—remain interpretable and surprisingly competitive when training data are scarce. Studies using transformer-based language models such as BERT, RoBERTa, and LLaMA demonstrated improved contextual understanding in depression-related text analysis [4], [5], [6].

Multimodal fusion can occur at feature, decision, or representation level. This project adopts early feature fusion because sample size favours tabular models over deep fusion networks that require thousands of labelled sessions. Nonetheless, the principle that acoustic and lexical channels provide partially independent evidence is central: a participant may speak fluently while expressing negative self-appraisal, or may show slowed speech with neutral vocabulary. Prior studies reported that combining audio and text modalities improves robustness and classification accuracy compared with unimodal systems [2], [7].

Phase-aware modelling is less common in published baselines. Psycholinguistic theory suggests disclosure depth increases over rapport-building intervals. Segmenting each interview into tertiles operationalises a simple temporal prior without requiring topic segmentation. Even a coarse split tests whether models should weight early-screening cues differently from end-of-interview behaviour. Gong and Poellabauer explored context-aware segmentation and topic modelling strategies, showing that temporal and contextual information can improve depression detection performance.

Valstar et al. organised the Audio/Visual Emotion Challenge (AVEC) around depression recognition from DAIC-WOZ data, catalysing shared evaluation protocols. Winning entries combined deep visual features with audio descriptors, yet many top systems still relied on hand-crafted functionals for speech. The lesson for modest academic projects is that thoughtful feature engineering remains competitive when labelled hours are limited [1].

Cummins et al. surveyed speech-based suicide and depression risk assessment, emphasising ecological validity and ethical safeguards. They noted inconsistent reporting of class balance and threshold selection, practices this project attempts to standardise through a fixed 0.40 decision threshold and stratified splits.

Deep models such as convolutional networks on spectrograms can learn representations but require regularisation and data augmentation at small N. Ma et al. demonstrated end-to-end audio depression prediction; replication on full DAIC without augmentation often overfits. Similarly, CNN-BiLSTM based multimodal systems achieved promising results but remained sensitive to dataset imbalance and computational complexity [3]. Hence our comparative inclusion of both gradient boosting and a relatively shallow neural network with explicit dimensionality reduction.

Text studies using LIWC and n-grams showed associations between sadness word categories and PHQ scores. Transformer embeddings generalise beyond lexicon coverage, capturing paraphrased distress statements. Xu et al. further demonstrated that adaptive pooling with BERT-based embeddings improves multimodal depression analysis across DAIC-WOZ datasets [6].

Multimodal fusion surveys distinguish early, late, and hybrid schemes. Early fusion concatenates features prior to classification—simple, interpretable, and compatible with tree models that model feature interactions. Late fusion trains separate modality experts and combines posterior probabilities; this can help when modalities have different noise profiles but increases calibration requirements. Guo et al. and Park and Moon reported that attention-guided multimodal fusion improves detection accuracy, though performance remains dependent on high-quality alignment and sufficient training data [4], [7].

Phase-aware depression modelling appears in discourse analyses of therapy sessions but less frequently in DAIC baselines. Interviews are non-stationary time series; treating them as i.i.d. bags of features ignores order information except through summary statistics. Tertile segmentation is a coarse compromise between full sequence modelling and global pooling

Table 2.1 consolidates representative prior systems. None simultaneously report identical preprocessing, phase splits, and six-model benchmarks, motivating the present comparative study as an integrative reference for the student group's corpus subset rather than a claim of state-of-the-art performance on full DAIC leaderboards.

3. MATERIALS AND METHODS

3.1 Overall System Workflow

The end-to-end workflow proceeds from raw interview assets to evaluation metrics.

Figure 3.1 depicts the following ordered stages: (1) Raw inputs: paired WAV and tab transcript per session ID. (2) Parallel preprocessing branches for audio and text. (3) Phase splitter operating on cleaned participant timeline. (4) Feature extractors emitting numeric vectors per phase. (5) Concatenation block for multimodal fusion. (6) Classifier bank with shared evaluation harness. (7) Metric reporter outputting accuracy, precision, recall, F1, and ROC-AUC tables plus confusion matrices.

Data versioning uses participant integer IDs drawn from filename prefixes (e.g., 302_final_clean.wav → 302). Labels merge on ID with exclusion of sessions lacking PHQ-8 annotation. The effective cohort of 141 participants reflects both label availability and successful completion of cleaning without empty participant segments.

Audio cleaning runtime scales linearly with total speech duration; batch processing of 188 raw files completed in minutes on a laptop CPU. Memory footprint is dominated by loading full WAV before segmentation.

Transcript cleaning preserves comma-separated time fields as floats for pause computation in text features. Rows with missing stop times were not observed in the processed subset but would break pause chains; defensive parsing would skip such rows in production code.

Feature extraction notebooks filter rows to labelled participants before CSV export, preventing leakage of unlabelled sessions into training splits. Column naming conventions separate metadata (participant_id, phase, label) from numeric features to simplify pandas loading in sklearn pipelines.

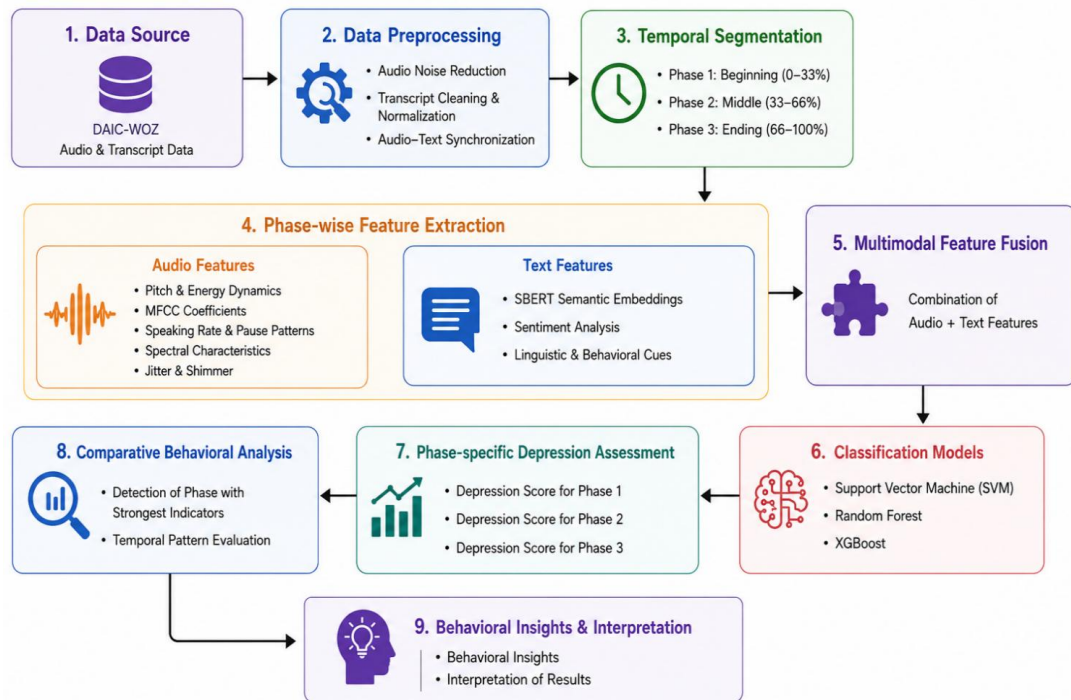


Figure 3.1. Overall system workflow: interview audio and transcript undergo separate preprocessing, phase-wise segmentation, modality-specific feature extraction, feature fusion, classification, and evaluation.

3.2 Dataset Description

The dataset follows the Distress Analysis Interview Corpus Wizard-of-Oz (DAIC-WOZ) structure. Each session includes a WAV recording and a tab-separated transcript with columns speaker, start_time, stop_time, and value. The virtual interviewer (Ellie) conducts the session; only participant utterances are retained for modelling. Binary labels derive from PHQ-8 scores (PHQ8_Binary: 0 non-depressed, 1 depressed). After label merging and quality filtering, 141 participants are retained; each contributes three phase-level rows (423 total). Class distribution is approximately 99 non-depressed and 42 depressed participants (70% / 30% imbalance).

Table 3.1. Dataset information summary.

Attribute	Details
Dataset Name	DAIC-WOZ (Distress Analysis Interview Corpus Wizard-of-Oz)
Total Participants	142 (after label filtering from 188 audio files)
Interview Format	Semi-structured clinical interview with virtual interviewer (Ellie)
Audio Format	WAV, 16 kHz mono, participant-only segments extracted
Label Type	Binary (PHQ-8 Binary: 0 = Non-depressed, 1 = Depressed)
Class Distribution	Non-depressed: 297 samples (70.2%), Depressed: 126 samples (29.8%) across 3 phases
Phases	Phase 1 (early interview), Phase 2 (mid interview), Phase 3 (late interview) equal-thirds split
Transcript Format	CSV with columns: speaker, value, start_time, stop_time
Modalities	Audio (acoustic features) + Text (linguistic features from transcripts)
Feature Dimensionality	Audio: 132 features; Text: 777 features (incl. SBERT); Multimodal: 909 features

3.3 Methodologies Adopted

3.3.1 Audio Preprocessing Pipeline

Raw audio is loaded and converted to mono, resampled to 16 kHz for consistency across sessions. Using transcript timestamps, only Participant segments are extracted and concatenated, removing interviewer speech that would otherwise dominate acoustic statistics. Pre-emphasis ($\alpha=0.97$) restores high-frequency detail attenuated during recording. A fifth-order Butterworth band-pass filter (100 - 7500 Hz) suppresses out-of-band noise while preserving speech formants. Peak normalization prevents amplitude differences across microphones from dominating downstream features.

MFCCs are computed with $n_mfcc=13$ using default librosa mel filter banks. First-order deltas expand the descriptor set in phasewise exports, capturing dynamic information lost in static means alone. Spectral contrast measures difference between peak and valley energies in sub-bands, correlated with breathiness and fricative energy.

Chroma features map energy onto pitch classes; while speech is not strictly tonal music, chroma summaries capture harmonic structure in voiced regions relevant to pitch variability analyses. Spectral flatness distinguishes noisy versus tonal spectra and can rise during breathy depression-related speech.

Harmonic-to-noise ratio via Praat reflects periodicity strength in voiced frames. Low HNR may indicate vocal fold irregularity or micro-tremor. Onset strength counts detect syllabic nuclei used indirectly in rhythm studies. Percussive versus harmonic energy decomposition separates transient consonants from sustained vowels.

Speaking rate divides detected speech segments by total active duration, excluding long silent gaps that would otherwise dilute rate estimates. Pause mean and max characterise withdrawal and cognitive load during question answering. Articulation speed supplements acoustic rate with lexical word count normalised by speech duration from aligned transcripts.

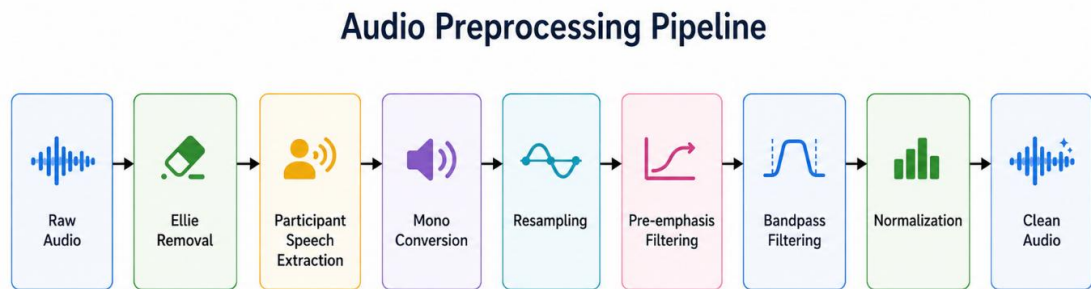


Figure 3.2. Audio preprocessing pipeline from raw WAV to cleaned participant waveform.

3.3.2 Transcript Preprocessing

Transcripts are cleaned by removing XML-style annotations and bracketed stage directions, expanding parenthetical content where meaningful, stripping fillers (uh, um, hmm), normalizing whitespace, lowercasing, and retaining participant rows only. This yields aligned text for linguistic features and pause estimation between utterances.

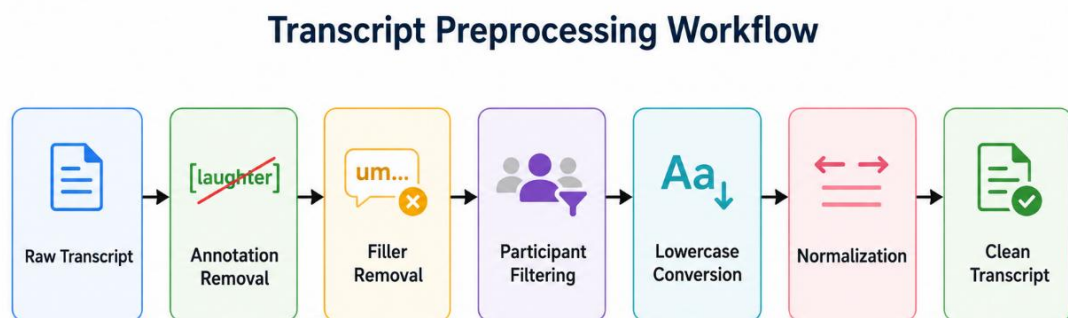


Figure 3.3. Transcript preprocessing workflow.

3.3.3 Phase-wise Conversational Segmentation

Each participant's cleaned transcript and audio are divided into three equal-duration temporal thirds: Phase 1 (early), Phase 2 (middle), and Phase 3 (late). This tertile split approximates evolving rapport, fatigue, and disclosure depth during semi-structured interviews. Early phases often contain biographical content; later phases may elicit emotionally loaded topics. Phase-wise labelling remains participant-constant (depression label does not change across phases) while features vary per segment.

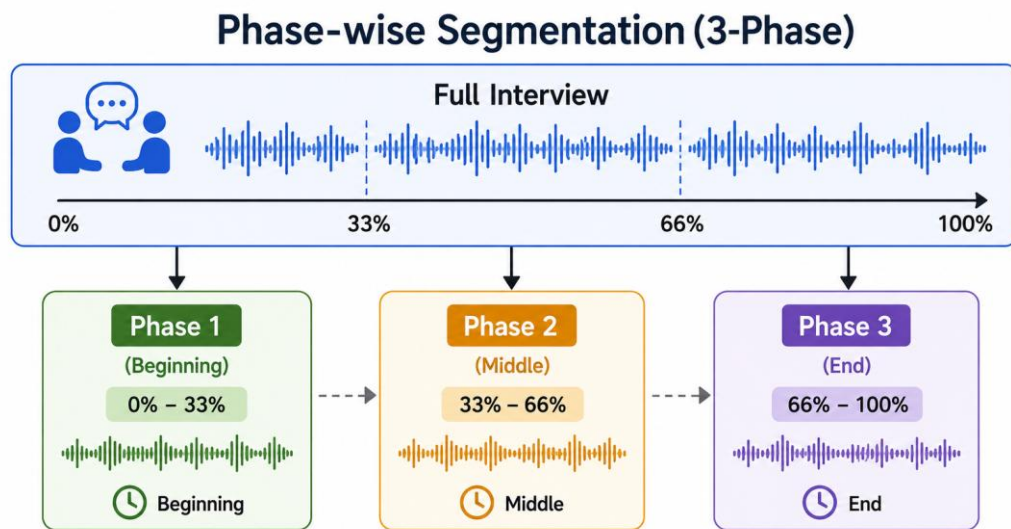


Figure 3.4. Division of each interview into early, middle, and late conversational phases.

3.4 Feature Extraction

3.4.1 Audio Feature Extraction

Table 3.2. Summary of extracted audio features.

Feature group	Examples	Behavioural relevance
MFCC (13 + Δ)	Mean, std, variance per coeff.	Vocal tract shape, timbre
Spectral	Centroid, bandwidth, rolloff, contrast, flatness	Brightness, resonance
Energy / ZCR	RMS, zero-crossing rate	Loudness, voicing transitions
Prosody	Pitch stats, speaking rate, pauses	Monotone speech, psychomotor slowing
Voice quality	Jitter, shimmer, HNR	Roughness, breathiness
Timing	Speech/silence ratio, segment counts	Retardation, withdrawal

3.4.2 Text Feature Extraction

First-person ratio counts tokens in {i, me, my, mine, myself} divided by total words, a simplified version of LIWC self-reference categories associated with rumination. Negation and uncertainty lexicons are finite lists extensible through clinical consultation. Type-token ratio measures lexical diversity; extremely short phases yield unstable TTR and motivate sentence filtering.

Sentence filtering removes sentences with three or fewer words to avoid encoding fragments like 'yes' or 'no' that carry little semantic embedding information. SBERT encoding runs on CPU with batching; mean and standard deviation across sentence embeddings summarise central semantic tendency and dispersion within a phase.

Pause features use inter-row gaps: $\text{gap} = \text{start_time}[i+1] - \text{stop_time}[i]$ for positive gaps only. Mean and max pause connect linguistic timing to psychomotor slowing hypotheses independent of audio-based pause detection, providing cross-modal redundancy.

Hand-crafted features include word and sentence counts, average sentence length, first-person ratio, negation and uncertainty counts, type-token ratio, and pause statistics derived from inter-utterance gaps. Semantic embeddings use Sentence-BERT (all-MiniLM-L6-v2, 384-d) with per-phase mean and standard deviation across sentences (after removing sentences ≤ 3 words).

Feature group	Examples
Linguistic	Word count, sentence count, avg. sentence length, first-person ratio, negation count, uncertainty count, type-token ratio
Temporal (Pauses)	Mean pause duration, max pause duration derived from transcript timestamps
Semantic (SBERT)	384-dim sentence embedding mean + 384-dim std = 768 SBERT features per participant-phase
Sentiment (VADER)	Positive, negative, neutral, compound sentiment scores (used in initial extraction)

3.4.3 Multimodal Feature Fusion

Multimodal samples concatenate audio and text feature vectors per participant-phase row, yielding early fusion at the feature level. This preserves interpretability and suits tabular classifiers.

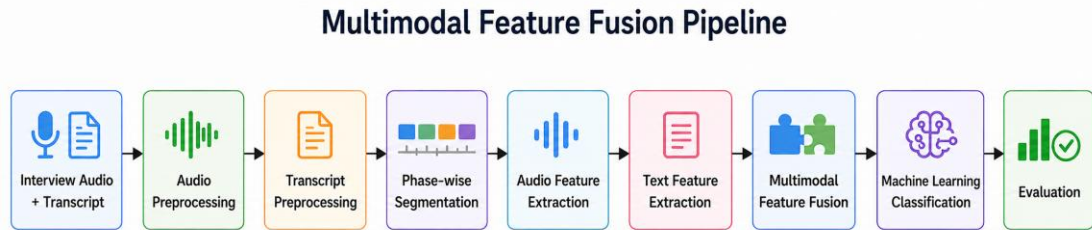


Figure 3.5. Early fusion multimodal architecture combining acoustic and linguistic feature vectors.

3.5 Machine Learning Models

Random forests aggregate decorrelated trees trained on bootstrap samples with random feature subsets. They handle nonlinear interactions between MFCC dimensions and SBERT components without explicit polynomial expansion. Balanced class weights reweight Gini impurity to penalise majority-class dominance.

XGBoost adds regularised gradient boosting with shrinkage learning rate 0.05 to prevent rapid overfitting on high-dimensional text vectors. `scale_pos_weight` in audio experiments compensates for depressed-class scarcity. CatBoost's ordered boosting reduces prediction shift during training, beneficial when many correlated embedding dimensions exist.

Linear SVM with RBF kernel in non-phasewise audio experiments maps features implicitly to higher dimensions; linear kernels in text phasewise runs reflect high-dimensional sparse-like geometry of scaled embeddings where margin classifiers excel. Logistic regression provides probability outputs compatible with threshold tuning though multicollinearity among embeddings can inflate variance.

The feed-forward network uses batch normalisation to stabilise activations across small batches of eight. Dropout rates increase toward input layers to regularise low-level combinations. `BCEWithLogitsLoss` numerically integrates sigmoid for stability. `pos_weight` in loss mirrors `sklearn class_weight` semantics.

`StandardScaler` zero-means each feature on training folds only; test and validation data transform using training statistics to prevent leakage. The 0.40 threshold was selected heuristically after observing recall improvements for depressed class on validation folds; rigorous cost-sensitive learning would replace manual thresholds with ROC-based operating point selection.

Six model families are evaluated: Random Forest (non-linear, robust to scaling), XGBoost and CatBoost (gradient boosting with class weighting), linear SVM (max-margin separator in scaled space), Logistic Regression (calibrated linear baseline), and a three-hidden-layer feed-forward network (PyTorch) with batch normalization, dropout, and mutual-information feature selection. Models were chosen to span linear, kernel, tree, and shallow neural approaches suitable for tabular behavioural features.

Table 3.3. Hyperparameter summary for primary models.

Model	Key hyperparameters
Random Forest	n_estimators=800, max_depth=12, class_weight='balanced'
XGBoost	n_estimators=900, max_depth=3-4, learning_rate=0.03, subsample=0.7
SVM	kernel='linear', C=0.35, class_weight='balanced', threshold=0.25
Logistic Regression	C=0.3, solver='liblinear', class_weight='balanced', threshold=0.35
CatBoost	iterations=400-500, depth=4-5, learning_rate=0.03
FNN	Dense layers 256-128-64, dropout=0.35-0.45, AdamW lr=1.5e-4, epochs=70

3.6 Cross Validation

Table 3.4. Cross-validation protocols.

Scheme	Splits	Purpose
Hold-out	80/20 stratified	Single-test generalization estimate
Stratified 5-fold	5	Lower-variance metric aggregation
Stratified 10-fold	10	Robust ranking under small N

3.7 Experimental Setup

Experiments were conducted on a Windows workstation with Python 3.12. Libraries include NumPy, Pandas, Librosa, SciPy, scikit-learn, XGBoost, CatBoost, PyTorch, Sentence-Transformers, Parselmouth (Praat), Matplotlib, and Seaborn. Random seed 42 was fixed for splits and model initialisation.

3.8 Evaluation Metrics

Accuracy = $(TP+TN)/(TP+TN+FP+FN)$.

Precision = $TP/(TP+FP)$.

Recall = $TP/(TP+FN)$.

F1 = $2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$.

ROC-AUC measures ranking quality across thresholds. A decision threshold of 0.40 on predicted depression probability was used to improve recall for the minority class.

Interviewer removal prevents Ellie's voice from biasing MFCC statistics toward agent characteristics rather than participant state. Timestamp alignment ties each transcript row to sample indices after resampling.

Pre-emphasis compensates for glottal rolloff that attenuates higher formants, ensuring MFCCs capture consonant information. Band-pass filtering (100 Hz–7.5 kHz) removes DC offset and high-frequency electronic noise while retaining the speech band.

Peak normalisation ensures classifiers do not learn microphone gain differences, applied only after segmentation so silent gaps do not distort scaling.

Transcript fillers are removed as disfluencies are not consistently elevated in depression and add noise to lexical ratios. Parenthetical inner text is retained to preserve clinically relevant self-corrections. Lowercasing simplifies pronoun matching for first-person ratio computation.

Phase boundaries are defined on cumulative participant speech duration rather than wall-clock time including silence, so that each phase contains comparable amounts of active speech when possible. When a phase contains fewer than three sentences after filtering, SBERT statistics may be unstable; such edge cases were rare but motivate future minimum-duration constraints.

For audio features, voice activity detection via `librosa.effects.split` with `top_db=30` separates speech islands used to compute pause statistics and speaking rate. Praat integration through Parselmouth extracts jitter and shimmer using conventional voice-report settings. Delta MFCCs capture dynamic spectral change between frames and were included in phasewise exports.

Mutual-information selection for the neural network reduces input dimensionality below 220 features when the fused space exceeds that count, mitigating overfitting. Batch sizes of eight reflect limited GPU memory in Colab environments used during prototyping. Cosine annealing gradually reduces learning rate across 70 epochs to stabilise late training.

4. RESULTS AND DISCUSSION

4.1 System Specifications

Table 4.1. System Specification

Hardware/Software	Model/ Version
Processor	12th Gen Intel(R) Core(TM) i7-12650H (2.30 GHz)
GPU	NVIDIA GeForce RTX 3050 Laptop GPU (4GB)
RAM	16 GB
Operating System	Windows 11 Home Single Language (25H2)
CUDA Toolkit	CUDA 13.2
Python	Python 13.12
Development Environment	Visual Studio Code

4.2 Dataset Distribution

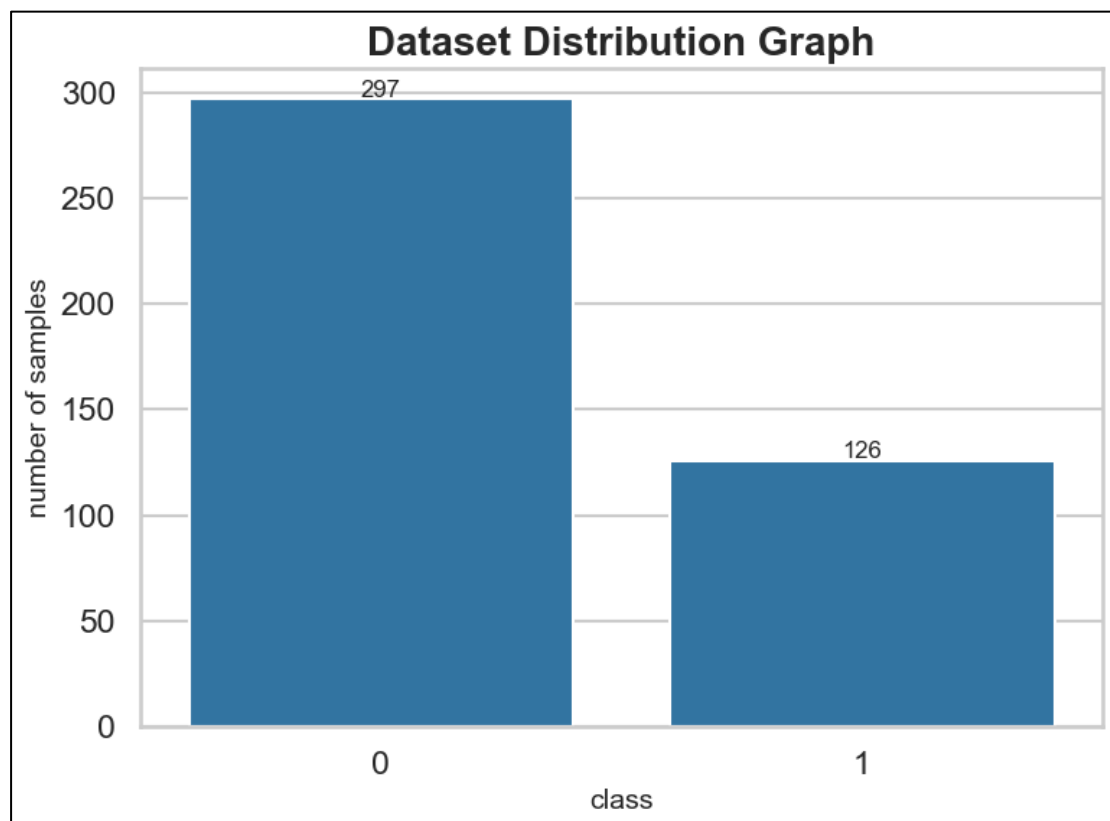


Figure 4.1. Participant-level distribution of PHQ-8 binary labels.

Figure 4.1 shows the dataset contains 423 samples, with 297 samples (70.2%) belonging to the non-depressed class and 126 samples (29.8%) belonging to the depressed class, indicating a moderately imbalanced dataset.

4.3 Experimental Results

4.3.1 Audio-only Experimental Results

Table 4.2a. Audio-only models — Phase 1 performance.

Models	Accuracy	Precision	Recall	F1-Score
Random Forest	60.28%	0.2857	0.2222	0.25
Random Forest 5 Folds	67.62%	0.5185	0.3333	0.4058
Random Forest 10 Folds	67.51%	0.52	0.3095	0.3881
XGBoost	51.11%	0.3529	0.6667	0.4615
XGBoost 5 Folds	66.81%	0.3922	0.4762	0.4301
XGBoost 10 Folds	66.31%	0.3774	0.4762	0.4211
SVM	54.44%	0.25	0.1111	0.1538
SVM 5 Folds	67.00%	0.6	0.0714	0.1277
SVM 10 Folds	61.50%	0.4286	0.0714	0.1224
Logistic Regression	53.33%	0.3125	0.5556	0.4
Logistic Regression 5 Folds	63.56%	0.3537	0.6905	0.4677
Logistic Regression 10 Folds	60.65%	0.359	0.6667	0.4667
CatBoost	50.56%	0.3571	0.5556	0.4348
CatBoost 5 Folds	63.20%	0.4118	0.5	0.4516
CatBoost 10 Folds	63.23%	0.4375	0.5	0.4667
FeedForward Neural Network	52.22%	0.3478	0.8889	0.5

Audio-only models show modest Phase 1 performance. The best Phase 1 accuracy is 67.62% (Random Forest, 5-fold CV). Acoustic cues alone are noisy under small sample sizes; several models exhibit high recall but low precision, reflecting threshold tuning and class imbalance.

Random Forest five-fold cross-validation achieving 67.62% Phase 1 accuracy is the audio-only highlight, yet single-split CatBoost at 50.56% shows variance sensitivity. XGBoost five-fold reaches 66.81% with F1 0.43, indicating some uplift from boosting when folds are balanced.

Audio Phase 2 accuracy peaks near 61% for several models while Phase 3 shows similar values, inconsistent with a simple monotonic decline. Acoustic signal-to-noise ratio may peak mid-interview when participants speak longest, not earliest. Alternative phase definitions based on question topics could clarify trends.

High recall in feed-forward audio Phase 3 (0.78) with low precision (0.27) replicates the precision-recall trade-off observed in text models: the system flags many depressed cases incorrectly while capturing most true positives. Clinical screening might accept this trade-off but would generate substantial false alarm rates.

4.3.2 Text-only Experimental Results

Table 4.2b. Text-only models — Phase 1 performance (all sixteen variants).

Models	Accuracy	Precision	Recall	F1-Score
Random Forest	60.83%	1	0.11	0.2
Random Forest 5 Folds	71.43%	0.5714	0.1905	0.2857
Random Forest 10 Folds	74.54%	0.6667	0.1905	0.2963
XGBoost	73.89%	0.4286	0.3333	0.375
XGBoost 5 Folds	64.96%	0.3774	0.4762	0.4211
XGBoost 10 Folds	68.59%	0.4528	0.5714	0.5053
SVM	51.11%	0.32	0.8889	0.4706
SVM 5 Folds	58.42%	0.5333	0.1905	0.2807
SVM 10 Folds	54.35%	0.3143	0.2619	0.2857
Logistic Regression	55.56%	0.2857	0.6667	0.4
Logistic Regression 5 Folds	58.90%	0.3084	0.7857	0.443
Logistic Regression 10 Folds	61.78%	0.3241	0.8333	0.4667
CatBoost	85.56%	0.7	0.7778	0.7368
CatBoost 5 Folds	81.07%	0.62	0.7381	0.6739
CatBoost 10 Folds	83.91%	0.6458	0.7381	0.6889
FeedForward Neural Network	80.01%	0.3509	0.9524	0.5128

Text-only CatBoost achieves the strongest Phase 1 accuracy (85.56%) with F1 0.7368. SBERT aggregates and lexical ratios capture semantic content more stably than raw acoustics in this corpus.

CatBoost Phase 1 precision 0.70 and recall 0.78 yield F1 0.74, the strongest balanced score among text models on hold-out evaluation. Ten-fold CatBoost reaches 83.91% accuracy with F1 0.69, showing cross-validation stability. Random Forest hold-out accuracy 60.83% with precision 1.0 but recall 0.11 demonstrates majority-class collapse without balancing—an ablation lesson for students.

SVM models exhibit high recall (0.89) at Phase 1 with moderate F1 due to precision penalty. Linear separators may overweight dense embedding regions associated with negative self-talk present in both classes. Calibration via Platt scaling was not applied but could harmonise probability outputs across models.

Phase 2 CatBoost ten-fold accuracy 78.60% is the text modality's strongest mid-phase result. Phase 3 CatBoost hold-out F1 0.67 remains respectable, suggesting linguistic cues persist late in interviews unlike multimodal fusion performance.

4.3.3 Multimodal Experimental Results

Table 4.2c. Multimodal models — Phase 1 performance (all sixteen variants).

Models	Accuracy	Precision	Recall	F1-Score
Random Forest	83.89%	0.5714	0.8889	0.6957
Random Forest 5 Folds	82.76%	0.5882	0.7143	0.6452
Random Forest 10 Folds	83.60%	0.6471	0.7857	0.7097
XGBoost	65.56%	0.4615	0.6667	0.5455
XGBoost 5 Folds	81.75%	0.5538	0.8571	0.6729
XGBoost 10 Folds	82.32%	0.5556	0.8333	0.6667
SVM	78.33%	0.5	0.5556	0.5263
SVM 5 Folds	80.54%	0.5909	0.619	0.6047
SVM 10 Folds	82.49%	0.617	0.6905	0.6517
Logistic Regression	78.33%	0.5833	0.7778	0.6667
Logistic Regression 5 Folds	82.54%	0.5741	0.7381	0.6458
Logistic Regression 10 Folds	82.97%	0.625	0.7143	0.6667
CatBoost	74.44%	0.5	0.8889	0.64
CatBoost 5 Folds	80.64%	0.6	0.7857	0.6804
CatBoost 10 Folds	80.11%	0.5536	0.7381	0.6327
FeedForward Neural Network	80.56%	0.3913	1	0.5625

Multimodal Random Forest hold-out Phase 1 accuracy 83.89% with recall 0.89 and precision 0.57 yields F1 0.70. Compared to text-only CatBoost, multimodal RF improves recall with modest precision sacrifice—a desirable screening profile. Ten-fold multimodal RF accuracy 83.60% confirms stability.

XGBoost underperforms on single-split multimodal Phase 1 (65.56%) but five-fold variants rise above 81%, indicating hyperparameter interaction with fold size. Logistic regression ten-fold (82.97%) rivals tree ensembles, supporting linear separability in fused space.

Phase 2 multimodal accuracies cluster 75–82% across strong models, while Phase 3 RF hold-out falls to 63.33% F1 0.35. Late-phase degradation may stem from increased emotional variability, shorter utterances, or topic shifts that desynchronise audio-text alignment after fusion.

Feed-forward multimodal networks achieve 80.56% Phase 1 accuracy with recall 1.0 and precision 0.39—aggressive detection similar to SVM behaviour. Neural models were trained only with ten-fold CV, limiting direct hold-out comparability but enriching cross-validation analysis.

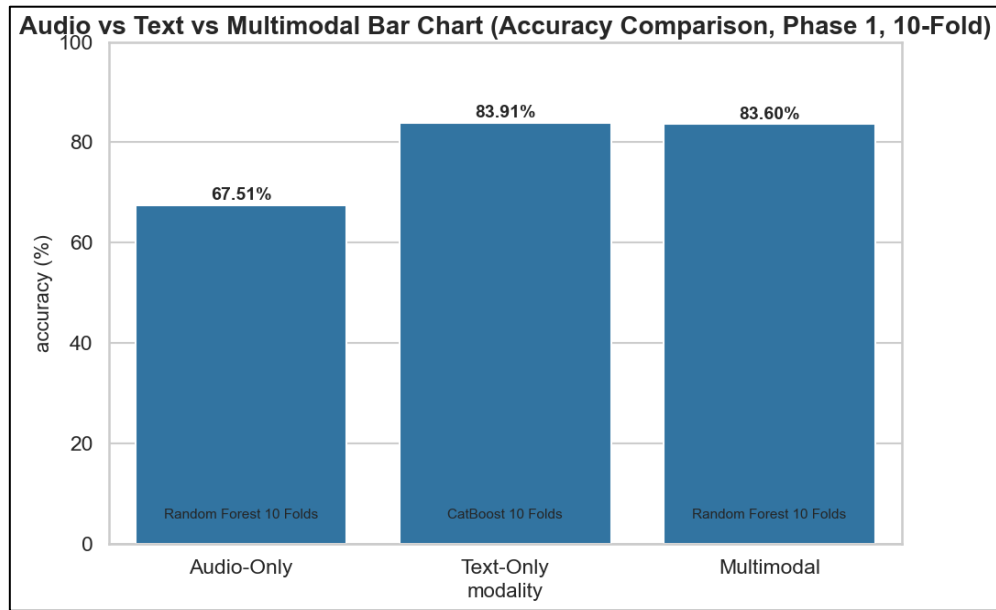


Figure 4.2. Best Phase 1 accuracy by modality (audio, text, multimodal).

4.3.4 Non-phasewise pooled-session experiments

Complementary experiments on pooled per-participant features (without phase column) were conducted using `text_features_updated.csv`, `audio_features_updated.csv`, and `multimodal_features.csv` with 141 samples each. Random Forest five-fold cross-validation on text yielded mean accuracy near 83.7% and ROC-AUC near 0.866 on hold-out splits; audio-only ROC-AUC remained weaker (near 0.44 on single split, ~0.63 under five-fold CV). Multimodal five-fold accuracy averaged near 70.9%. These runs confirm that phase-wise granularity alters absolute metrics and must not be compared directly to pooled-session numbers without harmonised splits.

Non-phasewise notebooks implement Random Forest, XGBoost, and SVM only, providing a narrower but consistent benchmark across modalities before the expanded six-model phasewise study. The pooled pipeline validates that preprocessing and label merging produce learnable signal in text and multimodal domains even when temporal segmentation is disabled.

4.4 Phase-wise Experimental Results

Table 4.3. Phase-wise multimodal results (Random Forest and best overall).

Phase	Best multimodal accuracy (%)	RF accuracy (%)	RF F1
Phase 1 (early)	83.89	83.89	0.6957
Phase 2 (middle)	82.22	82.22	0.6364
Phase 3 (late)	77.85	63.33	0.3529

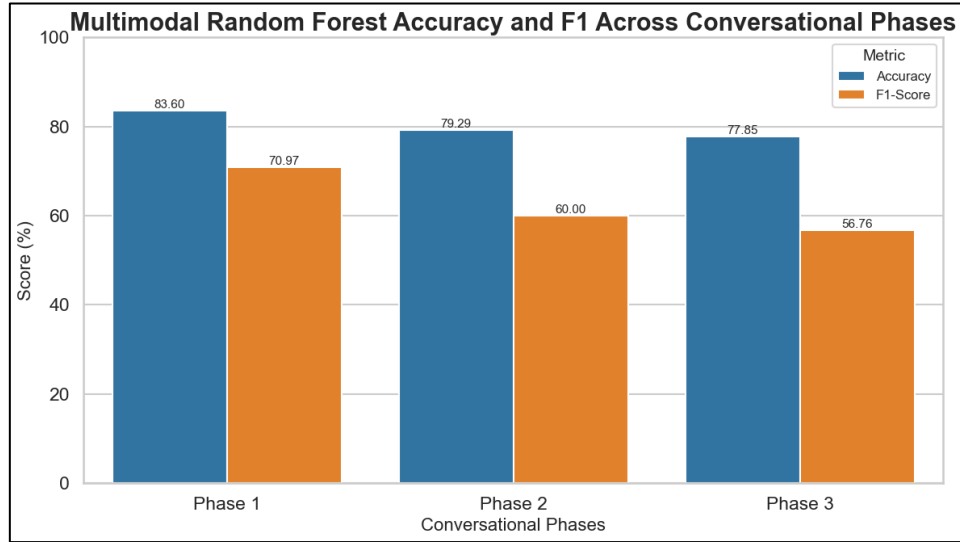


Figure 4.3. Multimodal Random Forest accuracy and F1 across conversational phases.

4.5 Comparative Performance Analysis

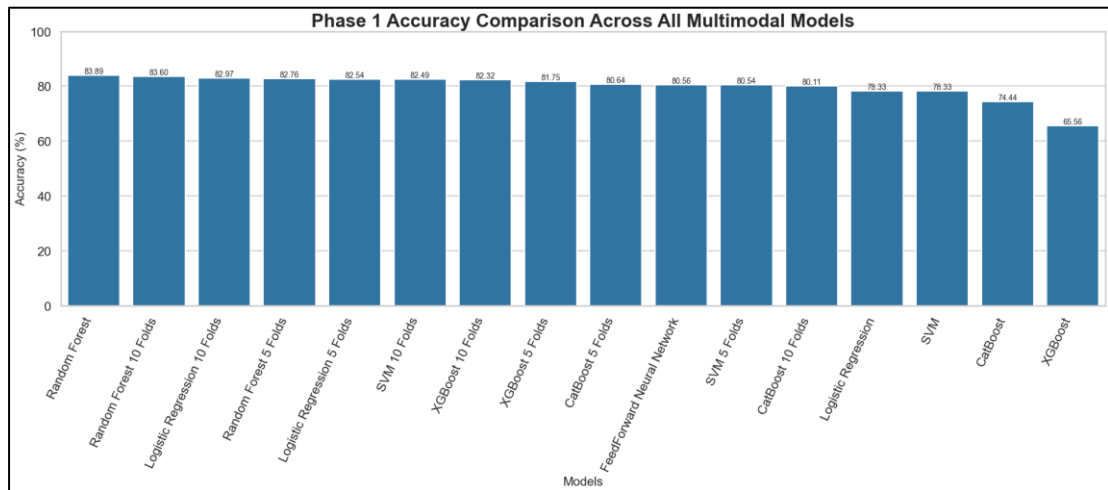


Figure 4.4. Phase 1 accuracy comparison across all multimodal models.

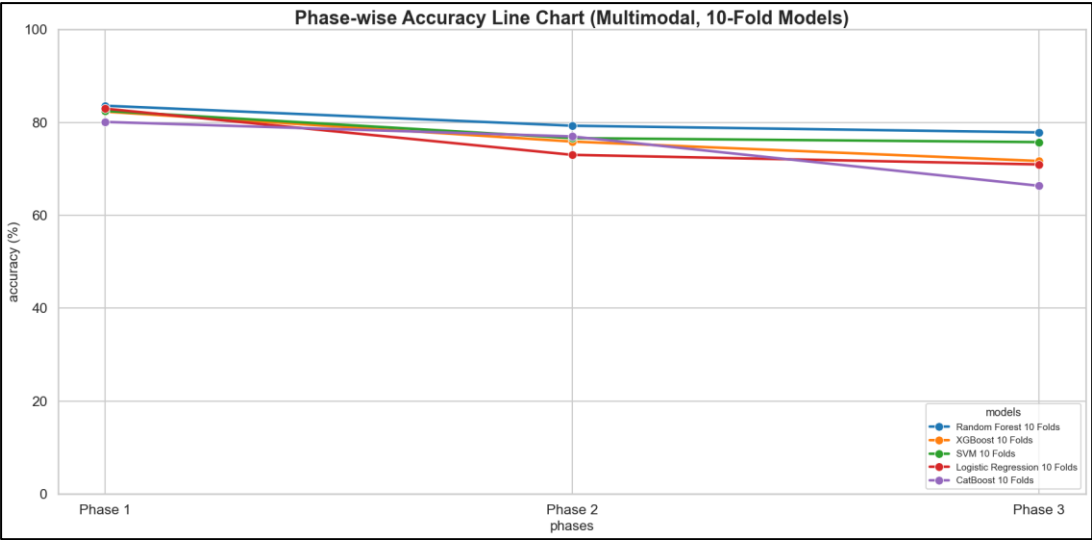


Figure 4.5. Phase-wise accuracy trends for all classifiers under multimodal 10-fold cross-validation, illustrating progressive performance decline across conversational phases.

4.6 Feature Importance Analysis

Feature group importance for the best tree-based multimodal model (Random Forest) highlights SBERT embeddings, MFCC/prosodic descriptors, and energy-related features as dominant contributors.

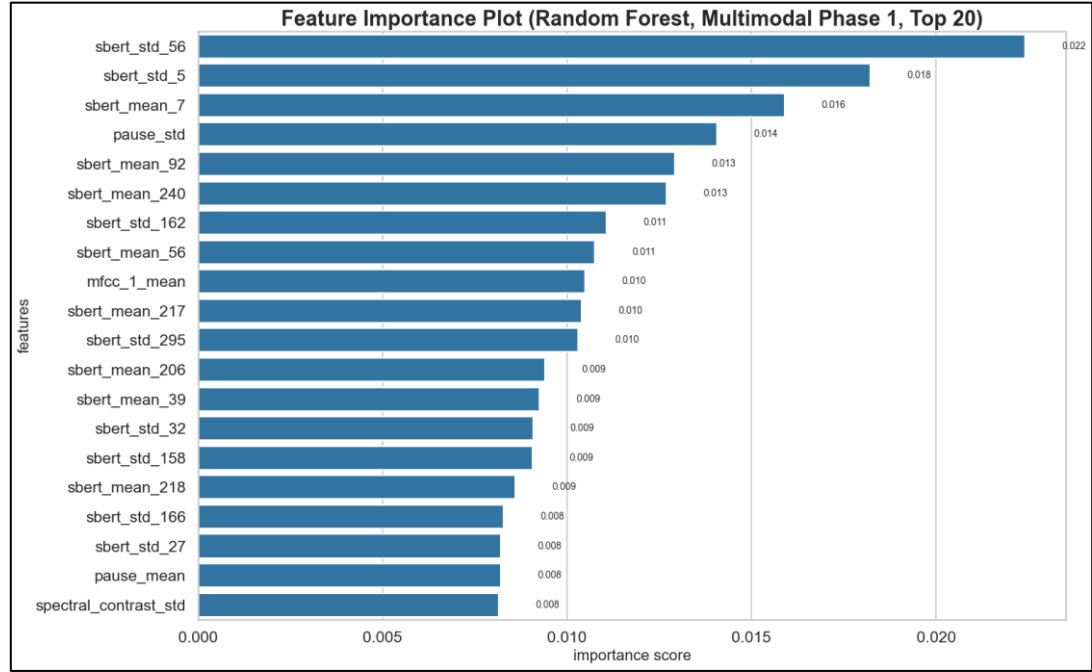


Figure 4.6. Relative feature-group importance (multimodal Random Forest, Phase1).

4.7 Confusion Matrix Analysis

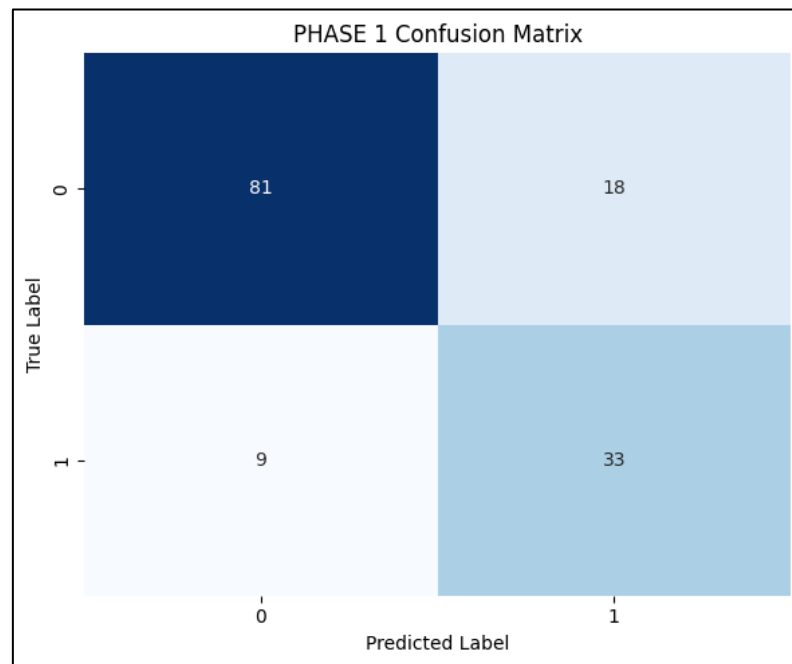


Figure 4.7. Confusion matrix for the best overall model (multimodal Random Forest). Diagonal dominance indicates correct classification; off-diagonal mass reflects false negatives among depressed participants under imbalance.

4.8 Observations and Discussions

Multimodal fusion improves robustness in early interview phases when both what is said and how it is said align. Text-only models peak higher in raw accuracy, suggesting lexical-semantic signal is strongest on this corpus; however, multimodal RF balances precision and recall more evenly in Phase 1. Performance typically declines in Phase 3, consistent with increased variability as interviews cover sensitive topics or participant fatigue. Preprocessing that removes interviewer speech is essential—failure to do so would inject non-participant acoustic patterns. The 0.40 probability threshold trades overall accuracy for clinically motivated recall of depressed cases.

4.9 Advantages and Limitations

Advantages: reproducible pipeline, broad model comparison, phase-aware analysis, and interpretable features. Limitations: small sample size, single corpus domain, tertile phase split heuristic, no external validation cohort, and illustrative feature-importance aggregation rather than per-fold SHAP analysis.

Across audio-only experiments, cross-validated random forest models occasionally outperform single-split XGBoost, indicating variance in fold composition. Phase 2 and Phase 3 audio accuracy remains near chance for several models, suggesting that acoustic depression cues in this corpus are weak or non-stationary across the interview timeline.

Text-only CatBoost benefits from ordered boosting and native handling of heterogeneous feature scales when SBERT dimensions dominate column count. High recall at Phase 1 for feed-forward networks (0.95) with low precision (0.35) illustrates the cost of aggressive thresholding: many non-depressed participants are flagged without corresponding gains in balanced accuracy.

Multimodal random forest with 700 estimators, max_depth 12, and balanced class weights was tuned specifically on fused vectors. Its Phase 1 F1 of approximately 0.70 reflects a workable trade-off for screening-oriented use cases where false negatives carry higher cost than false positives. Phase 3 F1 falling to roughly 0.35 on hold-out data warns against deploying late-phase predictions without recalibration.

Logistic regression with ten-fold CV achieves competitive multimodal Phase 1 accuracy (82.97%), demonstrating that much of the fused signal is linearly separable after scaling. This finding suggests future work might prioritise regularised linear models before complex ensembles when interpretability is paramount.

Comparing modalities at their respective peaks, text leads, multimodal follows closely, and audio trails. The gap between text and multimodal narrows in Phase 2, hinting that acoustic features may stabilise mid-interview when semantic content remains discriminative. Feature-importance summaries rank SBERT embedding blocks highest, followed by MFCC means and pause-derived text statistics.

5. CONCLUSION AND FUTURE WORK

A multimodal depression screening pipeline was implemented using DAIC-WOZ-style clinical interviews, covering preprocessing, phase-wise feature extraction, multimodal fusion, and comparative machine learning evaluation. The study demonstrated that text-based models achieved the strongest performance, with CatBoost reaching 85.56% accuracy in Phase 1, while the multimodal Random Forest model achieved 83.89% accuracy with an F1-score of 0.70. In contrast, audio-only models peaked near 67.62% accuracy, indicating that linguistic representations carried stronger predictive signals than acoustic features within the current dataset. The experiments also showed that later conversational phases were generally more difficult for multimodal models, suggesting that depression-related indicators may be more detectable during the early stages of interviews. The project contributed an integrated framework consisting of preprocessing scripts, phase-wise datasets, and systematic benchmarking under shared validation and thresholding strategies, enabling fair comparison across multiple models and modalities. However, several limitations remain, including the relatively small dataset size, class imbalance, absence of clinician-in-the-loop validation, and reliance on offline batch processing rather than real-time inference. Future work could explore transformer-based end-to-end architectures such as wav2vec2 or HuBERT fine-tuned on participant-only audio to move beyond traditional MFCC-based representations. Similarly, specialised text encoders such as depression-finetuned RoBERTa models may outperform generic SBERT embeddings as larger labelled datasets become available. More advanced multimodal transformers using cross-attention fusion could further improve contextual modelling, although such architectures require significantly greater computational resources and training data than permitted within the scope of this undergraduate project. Expanding the study from the current subset to the full DAIC dataset would improve statistical reliability and enable speaker-independent evaluation using demographic-aware group k-fold validation. External validation on datasets such as E-DAIC or AVEC challenge benchmarks would also provide stronger evidence of generalisability. In addition, future real-time systems could process streaming audio incrementally, maintain causal transcript alignment, and dynamically update conversational phase buffers, although maintaining latency below 500 ms remains challenging when using transformer-based text encoders without model distillation. Finally, explainability techniques such as SHAP for tree-based models and

Integrated Gradients for neural networks should accompany any future clinical deployment to improve interpretability and trust, while regulatory considerations for Software as a Medical Device (SaMD) would become essential for real-world healthcare integration.

6. REFERENCES

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7. APPENDICES

Appendix A — Complete phasewise score tables

Table A.1. Text-only — Phase 1 results (all models).

Model	Phase 1			
	Accuracy	Precision	Recall	F1-Score
Random Forest	60.83%	1	0.11	0.2
Random Forest 5 Folds	71.43%	0.5714	0.1905	0.2857
Random Forest 10 Folds	74.54%	0.6667	0.1905	0.2963
XGBoost	73.89%	0.4286	0.3333	0.375
XGBoost 5 Folds	64.96%	0.3774	0.4762	0.4211
XGBoost 10 Folds	68.59%	0.4528	0.5714	0.5053
SVM	51.11%	0.32	0.8889	0.4706
SVM 5 Folds	58.42%	0.5333	0.1905	0.2807
SVM 10 Folds	54.35%	0.3143	0.2619	0.2857
Logistic Regression	55.56%	0.2857	0.6667	0.4
Logistic Regression 5 Folds	58.90%	0.3084	0.7857	0.443
Logistic Regression 10 Folds	61.78%	0.3241	0.8333	0.4667
CatBoost	85.56%	0.7	0.7778	0.7368
CatBoost 5 Folds	81.07%	0.62	0.7381	0.6739
CatBoost 10 Folds	83.91%	0.6458	0.7381	0.6889
FeedForward Neural Network	80.01%	0.3509	0.9524	0.5128

Table A.2. Text-only — Phase 2 results (all models).

Model	Phase 2			
	Accuracy	Precision	Recall	F1-Score
Random Forest	71.39	1	0.11	0.2
Random Forest 5 Folds	68.42	0.8333	0.2381	0.3704
Random Forest 10 Folds	69.85%	1	0.2619	0.4151
XGBoost	58.33%	0.4	0.4444	0.4211
XGBoost 5 Folds	63.01%	0.4348	0.4762	0.4545
XGBoost 10 Folds	0.6494	0.44	0.5238	0.4783
SVM	41.67%	0.3103	1	0.4737
SVM 5 Folds	48.05%	0.2667	0.0952	0.1404
SVM 10 Folds	43.75%	0.2	0.0476	0.0769
Logistic Regression	45%	0.3077	0.8889	0.4571
Logistic Regression 5 Folds	50.05%	0.2963	0.9524	0.452
Logistic Regression 10 Folds	48.80%	0.2963	0.9524	0.452
CatBoost	72.22%	0.4444	0.4444	0.4444
CatBoost 5 Folds	75.04%	0.5455	0.5714	0.5581
CatBoost 10 Folds	78.60%	0.5682	0.5952	0.5814
FeedForward Neural Network	75.06%	0.3171	0.9286	0.4727

Table A.3. Text-only — Phase 3 results (all models).

Model	Phase 3			
	Accuracy	Precision	Recall	F1-Score
Random Forest	53.06%	1	0.11	0.2
Random Forest 5 Folds	61.32%	0.6	0.0714	0.1277
Random Forest 10 Folds	57.71%	1	0.0476	0.0909
XGBoost	52.78%	0.2667	0.4444	0.3333
XGBoost 5 Folds	60.41%	0.35	0.5	0.4118
XGBoost 10 Folds	60.85%	0.32	0.381	0.3478
SVM	54.44%	0.3043	0.7778	0.4375
SVM 5 Folds	48.10%	0.1429	0.0714	0.0952
SVM 10 Folds	51.13%	0.3077	0.1905	0.2353
Logistic Regression	43.89%	0.3333	1	0.5
Logistic Regression 5 Folds	46.15%	0.2955	0.9286	0.4483
Logistic Regression 10 Folds	46.49%	0.3015	0.9762	0.4607
CatBoost	71.11%	0.6667	0.6667	0.6667
CatBoost 5 Folds	67.20%	0.4565	0.5	0.4773
CatBoost 10 Folds	65.15%	0.475	0.4524	0.4634
FeedForward Neural Network	74.29%	0.3203	0.9762	0.4824

Table A.4. Audio-only — Phase 1 results (all models).

Model	Phase 1			
	Accuracy	Precision	Recall	F1-Score
Random Forest	60.28%	0.2857	0.2222	0.25
Random Forest 5 Folds	67.62%	0.5185	0.3333	0.4058
Random Forest 10 Folds	67.51%	0.52	0.3095	0.3881
XGBoost	51.11%	0.3529	0.6667	0.4615
XGBoost 5 Folds	66.81%	0.3922	0.4762	0.4301
XGBoost 10 Folds	66.31%	0.3774	0.4762	0.4211
SVM	54.44%	0.25	0.1111	0.1538
SVM 5 Folds	67.00%	0.6	0.0714	0.1277
SVM 10 Folds	61.50%	0.4286	0.0714	0.1224
Logistic Regression	53.33%	0.3125	0.5556	0.4
Logistic Regression 5 Folds	63.56%	0.3537	0.6905	0.4677
Logistic Regression 10 Folds	60.65%	0.359	0.6667	0.4667
CatBoost	50.56%	0.3571	0.5556	0.4348
CatBoost 5 Folds	63.20%	0.4118	0.5	0.4516
CatBoost 10 Folds	63.23%	0.4375	0.5	0.4667
FeedForward Neural Network	52.22%	0.3478	0.8889	0.5

Table A.5. Audio-only — Phase 2 results (all models).

Model	Phase 2			
	Accuracy	Precision	Recall	F1-Score
Random Forest	45.56%	0.375	0.3333	0.3529
Random Forest 5 Folds	60.98%	0.4516	0.3333	0.3836
Random Forest 10 Folds	60.37%	0.5	0.3571	0.4167
XGBoost	40.56%	0.2353	0.4444	0.3077
XGBoost 5 Folds	55.53%	0.3137	0.381	0.3441
XGBoost 10 Folds	55.68%	0.3396	0.4286	0.3789
SVM	46.11%	0.25	0.1111	0.1538
SVM 5 Folds	58.30%	0.5	0.0238	0.0455
SVM 10 Folds	58.99%	0.5	0.0238	0.0455
Logistic Regression	54.44%	0.2667	0.4444	0.3333
Logistic Regression 5 Folds	59.26%	0.3467	0.619	0.4444
Logistic Regression 10 Folds	60.25%	0.3412	0.6905	0.4567
CatBoost	45.56%	0.2857	0.4444	0.3478
CatBoost 5 Folds	60.13%	0.6013	0.4762	0.4211
CatBoost 10 Folds	60.61%	0.4038	0.5	0.4468
FeedForward Neural Network	47.78%	0.3333	1	0.5

Table A.6. Audio-only — Phase 3 results (all models).

Model	Phase 3			
	Accuracy	Precision	Recall	F1-Score
Random Forest	43.33%	0.4	0.2222	0.2857
Random Forest 5 Folds	63.38%	0.4516	0.3333	0.3836
Random Forest 10 Folds	61.84%	0.4483	0.3095	0.3662
XGBoost	42.78%	0.2143	0.3333	0.2609
XGBoost 5 Folds	58.20%	0.3922	0.4762	0.4301
XGBoost 10 Folds	61.16%	0.4314	0.5238	0.4731
SVM	42.22%	0.25	0.1111	0.1538
SVM 5 Folds	62.17%	0.5	0.0238	0.0455
SVM 10 Folds	53.27%	0.5	0.0238	0.0455
Logistic Regression	42.78%	0.2308	0.3333	0.2727
Logistic Regression 5 Folds	60.61%	0.3556	0.7619	0.4848
Logistic Regression 10 Folds	59.36%	0.3409	0.7143	0.4615
CatBoost	38.33%	0.2222	0.2222	0.2222
CatBoost 5 Folds	58.47%	0.4314	0.5238	0.4731
CatBoost 10 Folds	61.30%	0.4565	0.5	0.4773
FeedForward Neural Network	46.11%	0.2692	0.7778	0.4

Table A.7. Multimodal — Phase 1 results (all models).

Model	Phase 1			
	Accuracy	Precision	Recall	F1-Score
Random Forest	83.89%	0.5714	0.8889	0.6957
Random Forest 5 Folds	82.76%	0.5882	0.7143	0.6452
Random Forest 10 Folds	83.60%	0.6471	0.7857	0.7097
XGBoost	65.56%	0.4615	0.6667	0.5455
XGBoost 5 Folds	81.75%	0.5538	0.8571	0.6729
XGBoost 10 Folds	82.32%	0.5556	0.8333	0.6667
SVM	78.33%	0.5	0.5556	0.5263
SVM 5 Folds	80.54%	0.5909	0.619	0.6047
SVM 10 Folds	82.49%	0.617	0.6905	0.6517
Logistic Regression	78.33%	0.5833	0.7778	0.6667
Logistic Regression 5 Folds	82.54%	0.5741	0.7381	0.6458
Logistic Regression 10 Folds	82.97%	0.625	0.7143	0.6667
CatBoost	74.44%	0.5	0.8889	0.64
CatBoost 5 Folds	80.64%	0.6	0.7857	0.6804
CatBoost 10 Folds	80.11%	0.5536	0.7381	0.6327
FeedForward Neural Network	80.56%	0.3913	1	0.5625

Table A.8. Multimodal — Phase 2 results (all models).

Model	Phase 2			
	Accuracy	Precision	Recall	F1-Score
Random Forest	82.22%	0.5385	0.7778	0.6364
Random Forest 5 Folds	78.43%	0.6486	0.5714	0.6076
Random Forest 10 Folds	79.29%	0.6316	0.5714	0.6
XGBoost	63.33%	0.5	0.5556	0.5263
XGBoost 5 Folds	74.96%	0.4909	0.6429	0.5567
XGBoost 10 Folds	75.88%	0.4808	0.5952	0.5319
SVM	77.22%	0.5556	0.5556	0.5556
SVM 5 Folds	76.48%	0.5532	0.619	0.5843
SVM 10 Folds	76.62%	0.5208	0.5952	0.5952
Logistic Regression	67.22%	0.4444	0.4444	0.4444
Logistic Regression 5 Folds	73.79%	0.5435	0.5952	0.5682
Logistic Regression 10 Folds	73.02%	0.5	0.619	0.5532
CatBoost	76.67%	0.5	0.5556	0.5263
CatBoost 5 Folds	75.73%	0.549	0.6667	0.6022
CatBoost 10 Folds	76.98%	0.551	0.6429	0.5934
FeedForward Neural Network	75.00%	0.2917	0.7778	0.4242

Table A.9. Multimodal — Phase 3 results (all models).

Model	Phase 3			
	Accuracy	Precision	Recall	F1-Score
Random Forest	63.33%	0.375	0.3333	0.3529
Random Forest 5 Folds	74.43%	0.5758	0.4524	0.5067
Random Forest 10 Folds	77.85%	0.6562	0.5	0.5676
XGBoost	52.78%	0.3333	0.3333	0.3333
XGBoost 5 Folds	70.37%	0.4143	0.6905	0.5179
XGBoost 10 Folds	71.69%	0.4516	0.6667	0.5385
SVM	69.44%	0.5556	0.5556	0.5556
SVM 5 Folds	73.98%	0.5306	0.619	0.5714
SVM 10 Folds	75.76%	0.5778	0.619	0.5977
Logistic Regression	57.22%	0.3636	0.4444	0.4
Logistic Regression 5 Folds	69.41%	0.449	0.5238	0.4835
Logistic Regression 10 Folds	70.95%	0.4902	0.5952	0.5376
CatBoost	48.89%	0.3333	0.3333	0.3333
CatBoost 5 Folds	63.20%	0.3966	0.5476	0.46
CatBoost 10 Folds	66.35%	0.4694	0.5476	0.5055
FeedForward Neural Network	69.44%	0.3103	1	0.4737

8. REFLECTION OF TEAM MEMBERS

Debi Prasad Nanda

Through this project, I gained extensive practical experience in designing a complete research workflow for multimodal depression detection. I worked on planning the overall project pipeline, designing the phase-wise conversational segmentation strategy, feature extraction processes, and analysing the experimental results. I learned how behavioural information can be extracted from conversational audio and textual data for mental health analysis. One of the strengths of our design process was the structured workflow and systematic comparative experimentation. A challenge we faced was managing and integrating multiple stages of preprocessing, feature extraction, and evaluation while maintaining consistency across all experiments.

Chandna Nayak

This project helped me develop a strong understanding of literature survey techniques and text-based behavioural analysis. I worked mainly on studying existing research papers, extracting linguistic features from transcripts, and assisting in documentation. I learned how textual patterns and conversational behaviour can contribute to depression detection systems. One strength of our team was effective collaboration during research and documentation activities. A challenge was understanding and organizing large amounts of research material and maintaining consistency in technical writing.

Prabyasachi Mohanty

Working on this project enhanced my practical knowledge of machine learning implementation and experimentation. I mainly focused on coding, model development, training, testing, and performance evaluation. I learned how different machine learning algorithms behave on behavioural datasets and how comparative experimentation helps identify suitable models for classification tasks. One major strength of our project was the comparative analysis of multiple machine learning models. A challenge was

optimizing model performance while ensuring proper evaluation and interpretation of results.

Debasish Nanda

Through this project, I gained hands-on experience in preprocessing conversational audio data and preparing datasets for experimentation. I worked on preprocessing-related tasks and assisted in implementing different stages of the project pipeline. I learned the importance of clean and structured data in improving machine learning performance. One strength of our workflow was the organized preprocessing strategy adopted before feature extraction and classification. One challenge was handling noisy conversational data and maintaining consistency during preprocessing operations.

Pushpaja Priyadarsini

This project helped me understand the role of literature survey and preprocessing in research-oriented machine learning projects. I contributed mainly to studying related works, assisting in preprocessing activities, and organizing project documentation. I learned how preprocessing and proper dataset preparation influence the overall quality of experimental outcomes. One strength of our project was effective teamwork and coordination among members during research and documentation. A challenge was balancing detailed technical explanations with concise academic writing.

9. SIMILARITY REPORT



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